Topic 6: Energetics

Students will be assessed on their ability to:

6.1	know that the enthalpy change, ΔH, is the heat energy change measured at constant pressure and that standard conditions are 100 kPa and a specified temperature, usually 298 K
6.2	know that, by convention, exothermic reactions have a negative enthalpy change and endothermic reactions have a positive enthalpy change
6.3	be able to construct and interpret enthalpy level diagrams, showing exothermic and endothermic enthalpy changes
6.4	know the definition of standard enthalpy change of: i reaction, $\Delta_r H$ ii formation, $\Delta_f H$ iii combustion, $\Delta_c H$ iv neutralisation, $\Delta_{neut} H$ v atomisation, $\Delta_{at} H$
6.5	be able to use experimental data to calculate: i energy transferred in a reaction recalling and using the expression: energy transferred (J) = mass (g) × specific heat capacity (J g^{-1} $°C^{-1}$) × temperature change (°C) ii enthalpy change of the reaction in kJ mol^{-1} *This will be limited to experiments where substances are mixed in an insulated container and combustion experiments using a suitable calorimeter.*
6.6	know Hess's Law and be able to apply it to: i constructing enthalpy cycles ii calculating enthalpy changes of reaction using data provided, or data selected from a table or obtained from experiments
6.7	**CORE PRACTICAL 2** **Determination of the enthalpy change of a reaction using Hess's Law.**
6.8	be able to evaluate the results obtained from experiments and comment on sources of error and uncertainty and any assumptions made in the experiments *Students will need to consider experiments where substances are mixed in an insulated container and combustion experiments using, for example, a spirit burner and be able to draw suitable graphs and use cooling curve corrections.*
6.9	understand the terms 'bond enthalpy' and 'mean bond enthalpy', and be able to use bond enthalpies to calculate enthalpy changes, understanding the limitations of this method
6.10	be able to calculate mean bond enthalpies from enthalpy changes of reaction

	Further suggested practicals: i the enthalpy change for the decomposition of calcium carbonate using the enthalpy changes of reaction of calcium carbonate and calcium oxide with hydrochloric acid ii the enthalpy change of combustion of an alcohol iii the enthalpy change of the reaction between zinc and copper(II) sulfate solution iv the enthalpy of hydration of anhydrous copper(II) sulfate